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Choice architecture improves pension selection

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ABSTRACT

Consumers in Chile of private and semi-private pension systems are not selecting the pension providers that give them the largest Net Present Value. Regulators present retirees with a comparison report of pension plan offerings ranked by the amount offered and the risk classification of the provider. A field experiment implemented in Chile explores alternative presentations of provider performance. We treat respondents by modifying the status quo information 'package' (that exactly replicate what is currently provided by the regulator). Two design choices clearly enhanced consumer welfare in this experiment: Reducing the amount of information improved choice – specifically eliminating the risk profiles of providers leads to better decisions by our respondents. Second, adopting a loss frame also resulted in respondents selecting providers that generated higher returns. These gains from modifying information presentation were considerably higher for those with lower levels of financial literacy. This study is conducted in association with the *Superintendencia de Pensiones* (SP) and the *Comisión para el Mercado Financiero* (CMF), two of the public offices that oversee the pensions market in Chile.

KEYWORDS

Online Field Experiment;
Choice Architecture;
Pensions Decisions;
Behavioral Economics

Over the past couple of decades, research in behavioural economics has clearly established that the heuristics and biases that shape individual decision-making can lead to sub-optimal financial choices (Tversky and Kahneman 1974; Thaler and Sunstein 2009). Households are prone to make a variety of investment mistakes such as lack of diversification, risky share inertia, and a disposition to hold losing and sell winning stocks (Calvet, Campbell, and Sodini 2009; Balasuriya and Yang 2019). There is evidence of a U-shaped relationship between age and financial mistakes. Hastings, Madrian, and Skimmyhorn (2013) suggest, for example, that with age individuals accumulate increased financial decision-making abilities and then after the age of 50 these financial

reasoning abilities decline. A contributing factor to poor decision-making is low levels of financial literacy, which makes it difficult for the average consumer to understand complex financial issues (Lusardi and Mitchell 2007; Gathergood and Weber 2017).¹

In private or semi-private pension systems, where individuals contribute directly to a personal pension fund, they also get the opportunity to decide how to use that money once they retire. Some alternatives include lifelong Annuities, Lump Sum withdrawals, Programmed Withdrawals or some combination. The UK has multiple combinations of these alternatives, including a variety of nuanced variations in Annuities available to purchase.² So does the USA and

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¹See also the results of the global financial literacy survey <https://www.forbes.com/sites/maggiemcgrath/2015/11/18/in-a-global-test-of-financial-literacy-the-u-s/#5b10409358f0>, last accessed on 6 January 2018.

²See <https://www.ageuk.org.uk/information-advice/money-legal/pensions/what-you-can-do-with-your-pension-pot/or> <https://pensionsorter.co.uk/annuities/>, last accessed 13 April 2020.

Chile.³ This institutional design has the advantage of giving retirees the opportunity to choose – amongst a number of alternatives – for the best use of their retirement savings, given their personal circumstances. However, it requires retirees to make a long list of difficult decisions that have a direct impact on the level of financial security they will have for the rest of their lives. These are decisions that, for the most part, cannot be reversed. It is therefore not surprising that, in Chile, more than 80% of retirees end up paying large sums for delegating decisions to a pension advisor or Annuity sales agent, despite no guarantee that they will lead to the best possible outcome for one's pension (FNE 2018, 52).⁴

Recent contributions from the 'choice architecture' literature suggest that consumer welfare in the general area of consumer finance decision-making can be improved by enhancing the presentation of information associated with these choices. The U.S. CARD Act of 2009 (Soll, Keeney, and Larrick 2013) is a classic illustration and more recently Carpenter et al. (2020) demonstrate that enhanced design of information presentation for pre-paid cards can generate substantial consumer benefits. Similar welfare gains should be realized from the enhanced design of the information provided to retirees. We know choice architecture has a long-term impact on pension savings (Cronqvist, Thaler, and Yu 2018). Relatively minor information enhancements can significantly impact planning for Social Security (Liebman and Luttmer 2015). As Benartzi and Thaler (2007) note, there are no neutral choice architecture designs and even the small details could matter in the design of retirement saving vehicles.

However, we know much less about how choice architecture shapes one-shot-high-stakes decisions, such as what to do with the money accumulated in a defined contribution pension plan. Improving these high stakes, consumer decisions are a policy concern given the growing importance of defined contribution pensions such as U.S. Individual Retirement Accounts (IRAs) (Choi 2015). Providing 'full' information is not necessarily the

solution. The FNE (2018), for example, found that providing complete comparables on all pension alternatives did not lead consumers to make efficient choices FNE (2018).

We build on contributions from choice architecture and consumer decision-making and demonstrate that improving the design and presentation of comparative performance metrics causes significant welfare gains for consumers choosing amongst different pension providers. Heterogeneity in the treatment effects suggests that these welfare gains are realized disproportionately for the less financially literate who are more prevalent amongst the less well-off segments of the population.

Our causal evidence regarding this design architecture is based on an online experiment implemented in Chile. The treatment consists of versions of the actual pension offer report, generated by the *Sistema de Consultas y Ofertas de Montos de Pensión* (SCOMP).⁵ The SCOMP certificate is sent to customers during the retirement process. SCOMP is regulated by the Chilean pension authorities SP and CMF, partners in this study. We designed four alternative performance delivery metrics in addition to the existing version that was the control condition. Offer selection is incentivized, with participants earning more for selecting offers with higher net present value. Results suggest that reducing the amount of information in the cross-comparison tables and introducing a loss frame significantly improve offer selection in terms of the estimated net present value.

The online field experiment was conducted with the Centre for Experimental Social Sciences (CESS) Santiago's online subject pool from December 2018 to August 2019. The survey also collects information on the participant's risk preferences, knowledge of the pension providers, age, health conditions, and provides information about the pension system.

The article is organized as follows: The following sections provide summarized information on the Chilean pension system, for context, followed by

³See <https://www.spensiones.cl/portal/institucional/594/w3-article-3581.html> for information on Chile and <https://www.usa.gov/retirement> for the USA, last accessed 13 April 2020.

⁴The FNE is the Chilean National Economic Prosecutor's Office (*Fiscalía Nacional Económica*).

⁵SCOMP (<http://www.scomp.cl/>) use is mandatory when opting for a pension scheme. Its purpose is to provide complete and comparable information regarding life annuity offers and pension amounts in programmed withdrawals retirement alternatives, through a technological platform that connects AFPs, Life Insurance Companies and Pension Advisors.

literature on the impact of financial literacy, a detailed experimental design with each treatment, the specific treatment-related hypothesis, the data analysis and discussion.

I. Individual pension schemes

The Chilean defined contribution pension plan of 1980 requires all employed workers to contribute 10% of income to a personal pension pot, that is managed by a set of private companies denominated *Administradoras de Fondos de Pension* AFP (Pension Fund Administrator).⁶ These companies invest the individual's savings for a fee.⁷ At the end of a person's active employment period (in usual circumstances ages 65 for men and 60 for women), each individual can initiate the pension procedure, which starts with getting a report from the AFP with the total amount they have accumulated over time, which they then have to use to 'purchase' a pension scheme.

There are currently four variations of these pension schemes: 1) Annuity. Individuals buy the right to receive a fix and constant (inflation adjusted) amount from an insurance company. 2) Programmed withdrawals. A scheme that allows individuals to maintain ownership of the money that remains at the AFP and withdraws a proportion of it every month – with the financial and longevity risks associated. The maximum amount that is withdrawn is regulated by law and depends on the total funds available, the returns of the remaining savings and the yearly adjusted life expectancy, which in practice means the monthly amount decreases over time eventually getting to zero if the person lives long enough. 3) Sequential. There is also a sequential Programmed Withdrawals and Annuity scheme in which the customer starts off with a Programmed withdrawal, for a short and fixed time period, and then goes on to an Annuity. The Programmed Withdrawal periods provide a higher pension for the first years, but instead of going down gradually to zero, the pension levels off into an Annuity. To do this costumers split their funds into two pots, one for each pension scheme. 4) Combined.

There is a scheme that combines a simultaneous Programmed withdrawal and Annuity scheme. There is also a State guaranteed component called *Pensión Básica Solidaria* (PBS) of approximately 160 USD for individuals outside the defined contribution system, who have no retirement savings. For those who have some retirement saving, but their resulting pensions are low, there is a State subsidy called *Aportes Previsionales Solidarios* (APS). Both benefits are capped at individuals from the lowest 60% of the national income distribution. Individuals who have not accumulated sufficient funds to finance a pension at least equal to the PBS are not able to access the options described above.⁸

For any particular scheme, there are various pension providers and products. The product offerings are regulated by the government agencies, CMF and SP, and are implemented through a comparison website SCOMP intended to simplify the identification and comparison of offers made by different authorized providers. SCOMP produces a report with tables informing the offers for each of the pension schemes and special condition (nuanced variation) selected for a quote. The amount of information presented in each table in the report depends on the number of providers interested in sending a quote – which varies from 6 to 16. The number of tables depends on how many schemes (and their nuanced variations) the customer requests.

According to the Chilean National Economic Prosecutor's Office (*Fiscalía Nacional Económica* – FNE), consumer welfare is not being optimized: retirees opt, on average, for pensions that are 2% lower than what they could obtain from another provider with all other factors (e.g. pension scheme and guaranteed periods) held constant (FNE 2018). Their estimate is that 80% of all pension offers made between 2004 through to early 2017 experienced 'significant' losses (FNE 2018, 13). And the FNE concluded that a contributing factor could be the way pension offers are presented to retirees.

But selecting an optimal retirement offer is challenging for the average consumer, because it demands some degree of financial literacy, such as

⁶In 2019 this requirement was also compulsory for self-employed workers, even though contributing was optional since 2012.

⁷There are also voluntary contribution complements called which provide tax benefits for the extra savings.

⁸The information presented in this section is available in the Pensions Superintendency website (Superintendencia de Pensiones 2019).

understanding risk classifications. The choice overload literature suggests that a rich set of choices may not maximize consumer welfare (Schwartz 2004; Iyengar and Lepper 2001). The ‘paradox of choice’ here is the considerable number of vendors, in addition to the considerable amounts of information associated with each offer. There is evidence of welfare losses associated with this paradox of consumer finance-related choices (Carpenter et al. 2020). Although Scheibehenne, Greifeneder, and Todd (2010) make a convincing meta-analytic claim that the evidence of whether, and precisely how, this paradox of choice undermines consumer welfare is far from conclusive.

The ‘choice architecture’ literature suggests that the design of the decision-making context, specifically how choices are presented to a consumer, can reduce the welfare losses associated with this paradox of choices. Of particular relevance here is evidence that simple technical features of the visual display of multiple choices can improve the decision-making process (Samek et al. 2016). And there is a growing body of experimental evidence in the domain of consumer financial decision-making suggesting that regulating the content, amount and presentation of financial information to consumers can reduce welfare losses (Bertrand and Morse 2011; Carpenter et al. 2020). Moreover, there is also evidence that the welfare-enhancing benefits of choice architecture correlate with financial literacy (Agnew and Szykman 2005).

We contribute to this choice architecture literature with experimental evidence from the consumer finance domain, but specifically consumer choices regarding retirement offer selection. The online field experiment implemented in association with the Chile *Superintendencia de Pensiones* and *Comisión para el Mercado Financiero* identifies two design choices that improve consumer welfare: reducing the amount of information in the pension offer comparisons, and adopting a loss frame. We also provide some support for arguments that the effects of these design enhancements are correlated with financial literacy.

II. Experimental design

The study consists of an online field experiment that lasts approximately 30 minutes and is implemented through a combination of Qualtrics integrated with R through PHP.⁹ The user interface is in Qualtrics, using a simple and easy-to-read layout. Treatment assignment, implemented with R code, assigns participants to treatments that are tailored to the individual’s gender and socio-economic status. The treatments are designed to inform the choice architecture of pension comparison offers regulated by the SP and the CMF. Hence, we recruited female and male respondents between the ages of 55–70 who indicate they have not yet started their retirement process. This is the profile of individuals in the Chilean population who would typically make retirement provider choices similar to those in our decision-making treatments. The legal retirement age in Chile is 60 and 65 years old for women and men, respectively, and there are alternatives for early retirement (Superintendencia de Pensiones 2020).

Treatments are block random assigned by gender and income level; two primary sources of treatment effect heterogeneity (Gerber and Green 2012). The treatment decisions are incentivized with participants earning more for selecting offers with higher net present value. Together with the main treatment and control groups described below, the study includes financial literacy, self-reported health and mathematical ability and risk measures.¹⁰

In Module 1 of the experiment, we gather basic socio-demographics, including: gender, age, income, education, whether they have already applied for a pension, age at which their parents died, potential beneficiaries, knowledge about pension schemes and providers.¹¹ The information they provide on their gender and income groups is used to customize the pension offers presented to them in the Module 3 treatments.

As we described earlier, there are four different pension schemes into which Chileans can enrol. Respondents will be asked to make choices for two of these different pension schemes. Module 2 determines the two pension schemes for which respondents will be asked to make a decision.

⁹The experimental design and corresponding hypothesis were pre-registered with OSF <https://osf.io/ys5bk> while data collections was underway.

¹⁰The full questionnaire is available in the replication repository <https://github.com/deniselaroze/Experimento-Metricas-y-Formatos-SCOMP>.

¹¹If participants are not between 55–70 or have already obtained a pension, the experiments ends.

This happens in stages. First, participants are provided with basic information on the pension system, including descriptions of the four different pension schemes. This is done to ensure a basic level of common knowledge. Secondly, participants are presented with different retirement profiles and asked to indicate with which profile they most identify. Each participant is then asked to rank the four pension schemes in the order they believe is most relevant to the profile they had previously read and identified with (profile descriptions available in the Supplementary Annexe).¹²

Pension experimental vignettes

In Module 3 respondents are then presented with two decision-making vignettes corresponding to their two top-ranked pension schemes from Module 2. In each vignette, respondents are asked to select a pension provider. The offers for each vignette are presented in a separate table/figure (depending on treatment) that includes name of the company, monthly pension amount, risk classification of the company, net present value – variations depend on the treatments summarized in Table 1. Treatments are assigned *via* block randomization without replacement, so each participant gets to make decisions using two different information-delivery formats.

The information provided to participants in the treatments is real pension offers made to individuals of the same gender and income category as the

participant, between 2017 and mid 2018, and they include the true name of the company that made the offer.¹³ The format is consistent across schemes with the exception that risk classification is omitted for Programmed Withdrawals providers because these companies are not subject to risk classifications.

The offer selection is incentivized and participants are told that the amount of money they earn depends on how the offers are ranked by a group of experts, and that these experts based their rankings on the monetary value of the offer and the risk classification of the company (exact wording and payments in the Supplementary Annexe Fig. A.1). In practice, the experts of the SP, together with representatives of the CMF, used the Estimated Net Present Value (NPV) of the offers – using the formulas shared by experts in the FNE. This NPV estimates the expected lifetime payments from each offer and weighs them by the riskiness of the company, the guaranteed periods selected for the Annuity (if any) and the person's life expectancy.¹⁴ Offers that are ranked highest paid 1,500 CLP decreasing by 100–400 CLP intervals and reaching zero for the lowest ranked offering – ties paid the same. As each subject made choices for two different vignettes, respondents could earn up to 3,000 CLP (approx. 4.5 USD) in this Module.

Treatments The treatments are administered as images embedded in the Qualtrics screen. Respondents observed a table with a list of pension providers – pension fund administrators and

Table 1. Treatment variants for each pension scheme.

Scheme	Variable	Control	Treatment	Treatment	Treatment	Treatment
			1	2	3	4
Annuity & Sequential*	Columns	Two	Two	Two	Three	One
	Currency	UF	CLP	CLP	CLP/NPV	CLP/NPV
	Risk	Yes	Yes	In a link	–	–
	Loss Frame	–	–	Yes	Yes	Yes
Programmed Withdrawals**	Format	Table	Table	Table	Table	Figure
	Columns	One	One	Two	Three	One
	Currency	UF	CLP	CLP	CLP/NPV	CLP/NPV
	Risk	–	–	–	–	–
	Loss Frame	–	–	Yes	Yes	Yes
	Format	Table	Table	Table	Table	Figure

*The Combined scheme is excluded from the experiment, because there are not enough original (real) offers to generate all treatment combinations.

**Providers of programmed withdrawals schemes do not have a risk classification, that is the only substantive difference.

¹²The information included was obtained from the SP website (<https://www.spensiones.cl/portal/compendio/596/w3-propertyvalue-2816.html>, last accessed 26 February 2019) and revised by specialists from the SP and CMF. Details of the profiles included in the survey are available in the Supplementary Annexe.

¹³The median pension offer for the corresponding gender, income and scheme combination is used to avoid biasing towards a particular company. For consistency, all data observed by a given respondent comes from the same matched retiree.

¹⁴The FNE calculated the NPV for each treatment, using the same function as in their report.

Renta Vitalicia Inmediata

Opción	Razón Social	Pensión mensual en UF sin retiro de excedentes	Clasificación de Riesgo de la Compañía de Seguros*
1	RENTANACIONAL	7,24	BBB+
2	BTGPACTUALVIDA	7,21	A
3	PENTAVIDA	7,18	AA
4	METLIFE	7,18	AA+
5	PRINCIPAL	7,17	AA
6	SECURITY	7,11	AA-
7	OHIONATIONAL	7,07	AA
8	CONFUTURO	7,02	AA
9	BICEVIDA	7	AA+
10	SURA	6,98	AA
11	CONSORCIOVIDA	6,96	AA+

* Las categorías de Clasificación de Riesgo que permiten a las Compañías ofrecer Rentas Vitalicias, ordenadas de mejor a inferior clasificación, son las siguientes AAA (mejor clasificación), AA, A, BBB (inferior). Cada una de estas categorías puede tener subíndices "+" o "-", siendo el subíndice "+" mejor que el "-".

Figure 1. Control treatment for *Annuities and Sequential* scheme. Note: This is the current format employed by the regulator, SCOMP.

Retiro Programado

Monto de pensión mensual durante el primer año					
Opción 1	Opción 2	Opción 3	Opción 4	Opción 5	Opción 6
AFP HABITAT	AFP PROVIDA	AFP CUPRUM	AFP PLANVITAL	AFP CAPITAL	AFP MODELO
5,72 UF	5,7 UF	5,7 UF	5,7 UF	5,7 UF	5,7 UF

Figure 2. Control treatment for *Programmed Withdrawals* scheme. Note: This is the current format employed by the regulator, SCOMP. Note: Providers for this Programmed Withdrawal scheme do not have a risk classification.

Renta Temporal con Renta Vitalicia Diferida de 4 años

Opción	Razón Social	Pensión mensual en pesos†	Clasificación de Riesgo de la Compañía de Seguros*
1	PRINCIPAL	331.902	AA
2	BTG PACTUAL VIDA	331.086	A
3	OHIO NATIONAL	330.814	AA
4	PENTA VIDA	328.910	AA
5	CN LIFE	324.557	AA
6	METLIFE	321.564	AA+
7	CONSORCIO VIDA	320.748	AA+
8	SURA	319.660	AA
9	BICE VIDA	311.499	AA+

† Valor calculado en base a UF del día 03/08/2018.

* Las categorías de Clasificación de Riesgo que permiten a las Compañías ofrecer Rentas Vitalicias son las siguientes AAA (mejor clasificación), AA, A, BBB (inferior). Cada una de estas categorías puede tener subíndices "+" o "-", siendo el subíndice "+" mejor que el "-".

Figure 3. Treatment 1 for *Annuities and Sequential* scheme. Note: Earnings are expressed in actual Chilean currency (CLP) rather than the financial transaction unit (UF).

Retiro Programado

Opción	Razón Social	Monto de pensión mensual durante el primer año†
1	AFP HABITAT	180.352
2	AFP MODELO	179.809
3	AFP PROVIDA	179.809
4	AFP PLANVITAL	179.809
5	AFP CAPITAL	179.809
6	AFP CUPRUM	179.809

† Valor de UF en pesos al día 03/08/2018

Figure 4. Treatment for Programmed Withdrawals scheme. Note: In Programmed Withdrawals variation there is no risk classification information.

insurance companies, depending on the pension scheme selected. Figures 1–7 present each of the treatments. The information presented is real and tailored to the subject's demographics and pension scheme preferences. See Fig A.2 for a screenshot of the implementation. Respondents are asked to select a single provider from the list. There was no default provider and respondents are required to select an option before moving on to the next screen.

Our conjecture is that simplifying the presentation of information about retirement options will result in consumers making choices with higher NPV. The treatments are

designed to incorporate the following simplifications:

a) A change in currency from a banking monetary unit (UF) to the standard Chilean peso currency (CLP) should make the comparison much more accessible to the average consumer. The control condition employs the UF while all four Treatments employ the standard Chilean peso currency.

b) Reducing the amount of information consumers need to consider simplifies decision making. We remove the explicit risk metric from the comparisons table and insert it as a link at the bottom of the page in

Renta Temporal con Renta Vitalicia Diferida de 4 años

Opción	Razón Social	Pensión mensual en pesos†	Pérdida anual*
1	PRINCIPAL	331.902	0
2	BTG PACTUAL VIDA	331.086	-9.792
3	OHIO NATIONAL	330.814	-13.056
4	PENTA VIDA	328.910	-35.904
5	CN LIFE	324.557	-88.140
6	METLIFE	321.564	-124.056
7	CONSORCIO VIDA	320.748	-133.848
8	SURA	319.660	-146.904
9	BICE VIDA	311.499	-244.836

† Valor calculado en base a UF del día 03/08/2018.

* Monto que dejaría de ganar cada año de vida.

Figure 5. Treatment 2 for all (Annuities, Sequential and Programmed Withdrawal) pension schemes. Note: This treatments adds a loss column and eliminates the risk column replacing it for the Annuities and Sequential schemes, with the sentence: "Para obtener mayor información sobre la clasificación de riesgo de las compañías de seguro haga click "aquí" [for more information on the risk classification of insurance companies click here], which would take them to a screen that contained the risk classifications of all companies (see A.3 in Annex). Programmed Withdrawals are not informed with risk classifications and therefore did not see the links.

Renta Temporal con Renta Vitalicia Diferida de 4 años

Opción	Razón Social	Pensión mensual en pesos†	Valor total estimado a recibir (largo plazo)*	Pérdida total estimada**
1	PRINCIPAL	331.902	74.122.826	0
2	BTG PACTUAL VIDA	331.086	73.948.929	-173.898
3	OHIO NATIONAL	330.814	73.882.649	-240.177
4	PENTA VIDA	328.910	73.462.338	-660.488
5	CN LIFE	324.557	72.501.631	-1.621.195
6	METLIFE	321.564	71.861.587	-2.261.239
7	CONSORCIO VIDA	320.748	71.681.373	-2.441.453
8	SURA	319.660	71.420.839	-2.701.987
9	BICE VIDA	311.499	69.638.998	-4.483.829

† Valor calculado en base a UF del día 03/08/2018.

* Estimación del valor total de la oferta de pensión, considerando esperanza de vida, riesgo de quiebra de la compañía de seguros y la tasa de descuento de los períodos garantizados, si corresponde.

** Estimación del dinero que dejaría de percibir de no elegir la opción 1.

Figure 6. Treatment 3 for all pension schemes. Note: This treatment incorporates the estimated Net Present Value in a total and loss frame. The monthly payments column presents the real (not estimated) earnings in CLP just as T.2.

Renta Temporal con Renta Vitalicia Diferida de 4 años

Valor total estimado a recibir (largo plazo)

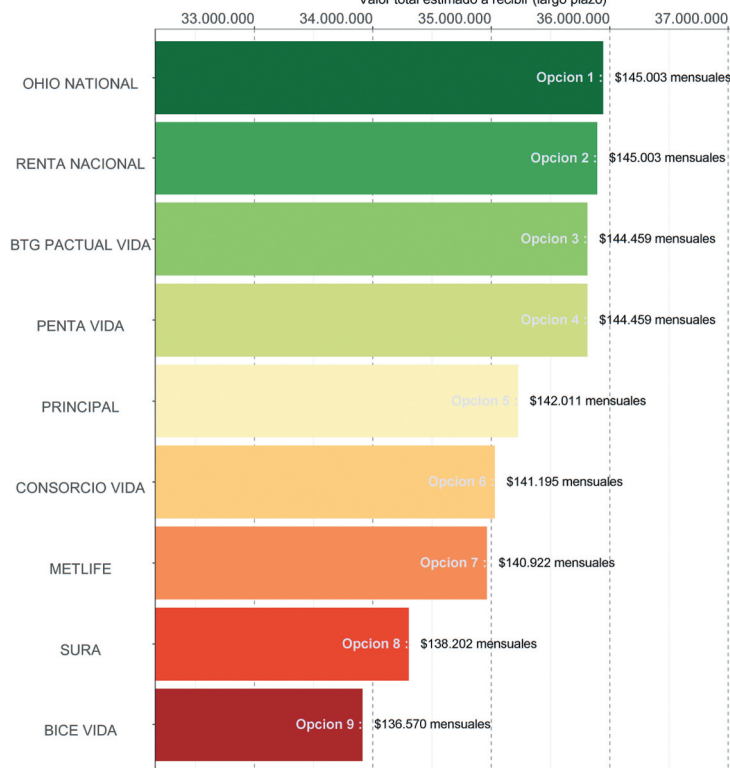


Figure 7. Treatment 4 for all pension schemes. Note: This treatment includes the same information as T.3 with a change in format. The loss framework is represented by the differences in the bars in the plot.

Treatment 2. Risk is directly incorporated into the comparative metric NPV in Treatments 3 and 4.

c) In Treatments 2 through 4, performance is expressed in a loss frame. The conjecture is that comparative performance expressed in a loss frame

would make it easier for consumers to select higher monetary offers.

d) The presentation format of offers should simplify offer selection. In Treatment 4 the offer comparisons are with a color coded graph (rather than a table).

Each row of [Table 1](#) summarizes the key dimensions of treatment variation: (a) the number of numeric columns, (b) the currency in which the offers are presented, (c) if there is a loss framework (d) the format in which the information is provided.

The first column of [Table 1](#) contains the presentation profile for the *control condition*. For all schemes, it uses the current SCOMP format as a baseline or control group. The *Treatment 1* in [Table 1](#) simply replaces the value of the offer in UF for the conversion to CLP. *Treatment 2* incorporates a *loss column*, that highlights the amount of money a person would lose in a year if they choose an offer different from the first. The treatment also removes the risk classification column and replaces it with a link (in [Figure 5](#) caption) to the risk classification of the insurance companies presented in Fig A.3 for Annuities and Sequential schemes. AFPs do not have risk classifications and therefore the link is not relevant for Programmed Withdrawal products. *Treatment 3* adds a total NPV and uses this metric in the loss framework, with losses over the entire life expectancy. By incorporating risk into the estimated NPV, this metric eliminates the need to present the risk classification as an independent value, without eliminating the impact of risk from the information provided to the retiree. *Treatment 4* presents the information in graphs with a green-yellow-red colour scale to highlight the most valuable options (in terms of estimated NPV). The bars in the plot represent the total estimated NPV, the monthly estimations are

incorporated next to the bars and vertical lines in the *plot that help participants visualize* the differences in the amounts offered.

The names of the companies that make the offers and amounts change for each gender, income and scheme group the respondent is categorized in, because the offers shown are specific for each group. This tailoring process is programmed so each participant can see offers that correspond with their gender and income group. The income levels used are: Level 1 – lowest 10% (217,000 CLP or less) of income with regards to the 2017 national register of active contributors to the pension fund; Level 2 – between 10% and 50% of income (217,001 CLP to 613,000 CLP), Level 3 – between 50% and 90% of income (613,001 CLP to 1,651,000 CLP); and Level 4 – the top 10% of income (1,651,001 CLP or more).¹⁵ Hence, each vignette will have offers tailored to the subject's gender and income level. [Table 2](#) summarizes these customized offerings.

After the two offer selection decisions, respondents enter Module 4 where they are asked a battery of questions regarding their risk and time preferences,¹⁶ self-perceptions of mathematical ability and complexity of financial services, standard financial literacy measures on compound interest rates, the bat-and-ball mathematical ability questions,¹⁷ and an incentivized one-question risk aversion measure (Eckel and Grossman 2008).

Heterogeneity Our information treatment is expected to affect the 'quality' (as NPV) of retirement offer selected by the subject. Respondents of course will differ in their facility with financial decision-making, and those with higher levels of facility with financial decision-making should, regardless of treatment condition, make better decisions. We include, as controls in estimating the treatment effects, three variables that measure facility. One of these is our measure of financial literacy that has been demonstrated to correlate with financial decision-making and in particular with retirement savings (Lusardi and Mitchell 2007; Lusardi 2012; Lusardi and Mitchell 2014; Hastings, Madrian, and Skimmyhorn 2013). A second is income, which we believe is likely to

Table 2. Mean offers in pesos for each of the four pension schemes.

Gender	Income level	Programmed withdrawals	Annuity	Sequential 2 years delay	Sequential 4 years delay
Female	217k <	162,943	154,648	139,234	135,455
	217k-613k	181,163	168,062	160,771	148,422
	613k-1,651k	251,836	238,429	228,416	226,883
	>1,651k	408,704	370,096	339,390	317,935
Male	217k <	182,871	175,438	157,060	159,547
	217k-613k	222,893	207,832	180,952	17,7879
	613k-1,651k	322,464	313,870	273,120	286,977
	>1,651k	717,134	691,026	606,617	574,311

¹⁵The percentiles 10 and 90 are chosen because they are used by the National Statistics Institute to measure income inequality and the 50th percentile is a simple mid point in the scale (INE 2017).

¹⁶Measured using the Spanish wording of the '2012 Global Preferences Survey elicit risk, time, and social preferences' (Falk et al. 2018).

¹⁷Re-phrased as a football and sneakers, as baseball is not a well-known sport in Chile and football is the national sport.

be correlated with the quality of decision-making. Our reasoning here is that those in higher income brackets will have more familiarity with the concepts used in the retirement products.

A third dimension is gender. There is evidence of gender differences in retirement investment decisions (Sundén and Surette 1998). The results of Lusardi and Mitchell (2008) and Lusardi and Mitchell (2014) lead us to expect male respondents to make better offer selections on average. Evidence suggests that men have higher levels of financial literacy than women. Hasler and Lusardi (2017) measure the financial literacy of Chilean adult women: 39% are categorized as financially literate compared to 41% of the adult population. It is also the case that in Chile men interact more with the pension system; 77.2% of men are active in the workforce and only 49.8% of women work outside the home. (INE 2018).¹⁸

Heterogeneous Treatment Effects. In our experiment, the average treatment effect is the difference in average returns from a simplified presentation of information versus those from the control status quo presentation. This treatment effect might differ for segments of the population and this difference is relevant for policy decisions. One possible source of heterogeneity is the subject's facility with financial decision-making. Our expectation is that this simplification of the presentation of performance information will have a higher effect for segments of the population with a lower facility with financial decision-making. The candidates for these heterogeneous treatment effects are the three we just described above: financial literacy, income and gender. We have no reason to believe that any particular one of these indicators will be the source of treatment effect heterogeneity. Hence, as we point out below, we employ a highly inductive machine learning technique to identify significant heterogeneous treatment effects.

Subject payments All respondents are paid 5,000 pesos for finishing the survey plus the amount they earn in each of the two pension offer tasks, and their earnings from the risk preference lottery. The payments on the pension offer tasks depend on the estimated NPV ranking of the offers for each specific

scheme and payments vary between 1,500 CLP for the offer with the highest estimated NPV, decreasing to 1,400 CLP, 1,200 CLP, 900 CLP, 500 CLP, 100 CLP and 50 CLP for the seventh ranked alternative, and 0 CLP after that. Fig. A.1 in the Supplementary Annexe presents the payment table shown to respondents and the exact wording of the instructions. This uneven ranking is intended to resemble the (usually) small monetary differences between the first few offers, reduces punishment for risk aversion at the top of the scale, and disincentivizes selecting offers in the lower end of the estimated NPV ranking. All rankings are constructed within each pension scheme and both selection tasks are independent.¹⁹ Total payoffs can vary between 5,000 CLP and 10,160 CLP, depending on the decisions made and luck. Respondents are informed of their total payment at the end of the experiment.

Subject recruitment A sample of 1041 respondents were recruited through the Centre for Experimental Social Sciences (CESS) Santiago online subject pool (<https://cess.cl/>). The CESS Santiago team conducted special recruitment campaigns using social media (Facebook, Twitter, Instagram), email campaigns and face-to-face recruitment and a publication in *Le Monde Diplomatique Chile*.²⁰

Summary of conjectures. Our central claim is that choice architecture, in this case the design of information regarding pension offers, can enhance consumer welfare. We expect the status quo, the Control Treatment, will result in the lowest percentage of maximal return offers being selected (measured in NPV) – which is equivalent to the lowest average experimental payoffs. The four other treatments should result in better choices. Treatment 4 should be particularly high, given the graphical presentation of offers.

Second, we expect heterogeneity in financial decision-making. We expect better financial choices from respondents with higher levels of financial literacy, with more income, and who are male. Accordingly, these are included as controls in estimating the treatment effects. Similarly, we expect treatment effects to be significantly higher for

¹⁸This was our pre-registered expectation, we do not find evidence of gender differences in our results.

¹⁹The instructions are transparent about payments being based on criteria defined by a group of experts. Those experts are representatives of the *Superintendencia de Pensiones* and the *Comisión para el Mercado Financiero*, that used the criteria of NPV as defined and calculated by the *Fiscalía Nacional Económica* (FNE 2018) – the main Chilean regulatory and oversight agencies.

²⁰In our pre-analysis plan, we had set out to collect 1000 observations. The extra 41 people was simply due to slightly over shooting the invitations.

respondents who are lacking in a facility for making financial decisions. Again, all three of these variables – gender, financial literacy and income – are candidates for conditioning the treatment effects.

III. Results

Our subject recruitment strategy targeted men and women with earnings in the middle- to high-income category; individuals who are active in the workforce and contributing to their personal pension account; and those who had not started the retirement process.²¹ In Chile, the legal retirement age is 60 years old for women and 65 for men. Given early retirement possibilities and late retirement decisions, our age threshold varies by ± 5 years; hence, it ranged between 55 and 70 years of age.²² This is the age range of individuals in the Chile population who are expected to make retirement provider choices similar to those in our treatments. The resulting sample is similar to this 55–70 age target population: it is predominantly male (56%), with 75% of the respondents between 55 and 63 (and a long tail that ends in 71); and 12% of the sample is in the lowest income group, 29% are level 2, 34% are level 3 and 25% are in the highest income group.

Table 3 presents the multi-nomial logit estimates from regressing treatment assignment on socio-demographic variables, risk preferences and financial literacy.²³ The regression coefficients on these treatment assignment predictor variables are not statistically significant suggesting no imbalance in treatment assignment. There is no evidence that gender, socioeconomic status, age or risk preferences are significant predictors of treatment assignment.²⁴ The sole exception here is the lower likelihood of observing treatment 4 relative to control for people that have a low level of financial literacy, relative to no financial literacy, and controlling for this does not alter statistical results. These are variables that are likely to be correlated with financial decision-making – hence any treatment assignment that is predicted by these variables would signal problems with the random assignment mechanism and undermine any causal

claims. This is not the case. The general balance of experimental treatment allocation that we find here provides confidence in a causal interpretation of the estimated treatment effects.

Treatment effects

The outcome of interest is the monetary gain associated with a subject's retirement product choice. Figure 8 presents the average returns for respondents assigned to Control and Treatments 1 through 4. Average returns for respondents assigned to Treatment 1 – only changing the currency – are not statistically different from those in Control. Treatments 2 through 4 reduce the amount of information (in particular, eliminating the risk classification variable from the comparables table) and implement a loss frame.²⁵ These two modifications in the presentation of performance information had a statistically significant effect on the respondents'

Table 3. Multinomial logit models on treatment assignment by socio-demographic characteristics – balance test.

	T.1	T.2	T.3	T.4
	(1)	(2)	(3)	(4)
Age:61–65	– 0.014 (0.156)	– 0.189 (0.152)	– 0.028 (0.151)	0.102 (0.154)
66+	0.069 (0.280)	0.241 (0.258)	0.138 (0.265)	0.169 (0.274)
Male	– 0.051 (0.150)	0.034 (0.143)	0.221 (0.145)	0.028 (0.148)
SEC 2	0.184 (0.252)	0.190 (0.241)	0.006 (0.238)	0.123 (0.244)
SEC 3	0.019 (0.249)	0.122 (0.236)	– 0.163 (0.234)	– 0.067 (0.241)
SEC 4	0.130 (0.265)	0.128 (0.254)	– 0.007 (0.249)	0.037 (0.258)
Risk level 2	– 0.066 (0.199)	0.088 (0.195)	0.154 (0.195)	0.117 (0.201)
Risk level 3	– 0.155 (0.227)	0.067 (0.219)	– 0.008 (0.222)	0.027 (0.228)
Risk level 4	– 0.514 (0.286)	– 0.155 (0.265)	– 0.044 (0.261)	– 0.129 (0.274)
Risk level 5	– 0.163 (0.213)	0.093 (0.206)	– 0.048 (0.210)	0.085 (0.212)
Low Fin. Lit.	0.007 (0.189)	– 0.298 (0.175)	– 0.248 (0.176)	– 0.360* (0.180)
Mid Fin. Lit.	– 0.006 (0.224)	– 0.178 (0.209)	– 0.341 (0.214)	– 0.279 (0.215)
High Fin. Lit.	0.258 (0.328)	– 0.286 (0.328)	– 0.346 (0.332)	– 0.227 (0.330)
Constant	– 0.038 (0.291)	0.155 (0.277)	0.184 (0.275)	0.092 (0.282)
Akaike Inf. Crit.	6,774.775	6,774.775	6,774.775	6,774.775
Note:	*p < 0.05; **p < 0.01; ***p < 0.001			

²¹Pensions for individuals in the middle- to low-income categories are organized differently through State benefit pensions.

²²For descriptive results see figures A.6 – A.8 in the Supplementary Annexe.

²³Table A.1 in the Annexe presents the number and proportion of participants that observed each treatment, by socio-demographic characteristic.

²⁴An age categorization is used to test balance due to the long tail of the age distribution. The estimation of treatment effects are not altered if one uses a continuous age variable or a categorization.

²⁵T.2 puts the risk classification in a link that interested parties can choose to click. T.3–4 incorporate risk in the NPV metric.

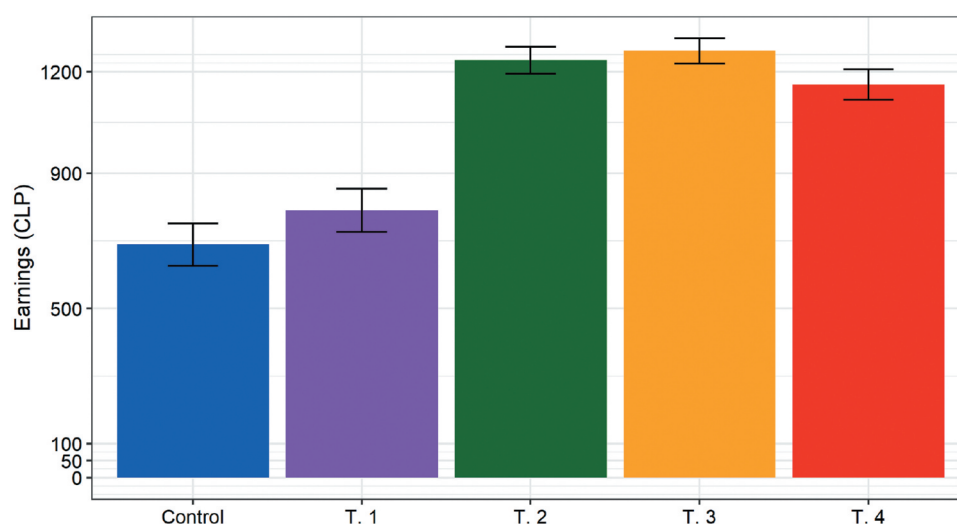


Figure 8. Experimental earnings by treatment.

average monetary gains. The average experimental gains in the Control and Treatment 1 conditions

Table 4. OLS estimations on amount earned in offer selection with participant-clustered standard errors.

	(1)	(2)
T.1	101.654* (45.476)	100.405* (45.060)
T.2	546.069*** (37.783)	543.776*** (37.683)
T.3	572.957*** (36.922)	570.121*** (36.780)
T.4	473.707*** (39.113)	468.485*** (38.967)
Age		2.173 (3.189)
Male		35.228 (23.731)
SEC 2		0.086 (37.813)
SEC 3		− 62.822 (38.190)
SEC 4		22.535 (39.656)
Risk level 2		− 18.518 (32.308)
Risk level 3		− 94.253* (38.533)
Risk level 4		24.371 (42.120)
Risk level 5		68.943* (33.581)
Low Fin. Lit		− 44.274 (27.993)
Mid Fin. Lit.		29.643 (34.060)
High Fin. Lit.		46.310 (51.651)
Constant	688.293*** (31.865)	− 3,553.932 (6,246.233)
N	2082	2082
Adjusted R ²	0.175	0.191

Note: *p < 0.05; **p < 0.01; ***p < 0.001.

ranged between 700 and 800 CLP – no statistically significant difference between the two. On the other hand, the average earnings for products chosen by respondents in Treatments 2 through 4 were roughly 1,200 CLP. And the difference between these average earnings and those in the Control and Treatment 1 conditions are statistically significant at the 95% level.

The use of a strong numeric loss treatment, in combination with the elimination of risk classification information, has a substantive impact on participants' propensity to select the offers with higher NPV (the metric suggested by the FNE (2018) report as the appropriate measure of comparison).

Table 4, presents the estimated OLS coefficients (with clustered standard errors) from the regression of offer selection earnings on Treatments; Model 2 includes socio-demographic controls. The Treatment effects are similar in the two models. There are large significant effects for Treatments 2 through 4 and a smaller, but statistically significant, coefficient for Treatment 1.²⁶ These Treatment effects have similar magnitudes and Model 2 results suggest the socio-demographic variables are not significantly correlated with the earnings for offer selection.

A second metric for assessing respondents' decisions is whether they selected the offer with the highest NPV (rather than our measure of offer earnings as above). Hence, the outcome of interest is whether participants selected the offer with the highest NPV. Figure 9 summarizes the results and

²⁶Standard errors are clustered at the individual level because each individual is randomly assigned two different pension schemes for which they make a decision.

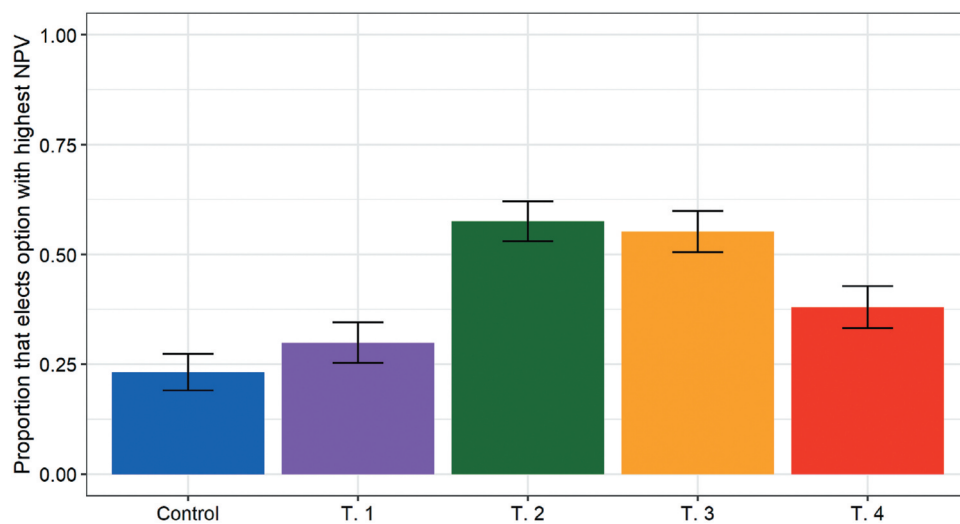


Figure 9. Proportion of the sample that selected the offer with the highest NPV per treatment.

they look very similar to those in Figure 8: In Control and Treatment 1 approximately 25% of the respondents select the offer with the highest NPV. In Treatments 2 and 3, a much higher proportion – over 50% – selected the highest NPV. In Table 5, Model 1 presents the logistic regression results. Again, as is the case in Table 4, Treatments 2 through 4 exhibited strong effects compared to Control while Treatment 1 had a much smaller coefficient. Treatments 2 and 3, that add a loss frame and eliminate risk classifications, have the strongest impacts, with twice the coefficient for Treatment 4 and five times as much as Treatment

1. As before, these findings suggest a strong impact of a clear loss frame, in combination with the elimination of a visible risk classification – the NPV incorporates risk into the metric eliminating the A+ type categories.

Based on our reported treatment effects we are very confident in concluding that some combination of reducing information and adopting a loss frame improves consumer welfare in this particular case. But Treatments 2 through 4 incorporate both a loss framework and eliminate the risk classification from the offer comparison table at the same time, making it difficult to identify the effect of each element. We can isolate the individual effect of the loss framework by focusing only on choices related to Programmed Withdrawals because providers of these products do not have a risk classification. For the Programmed Withdrawals, the estimated treatment effects only contain the impact of the loss frame. Models 2 and 3 in Table 5 present results only for the Programmed Withdrawal decisions. In the case of the Programmed Withdrawals, Treatment 1 serves as the control condition.²⁷ Model 2 presents the difference between Treatments 2 through 4, relative to Treatment 1 for the full sample. Model 3 presents the same estimations for the Programmed Withdrawals decisions hence a subset of the observations in Model 2. The magnitude of the coefficients in Tables 2 and 3 is

Table 5. Logit estimations on maximal offer selection with participant bootstrap-clustered standard errors. Model 1 includes all treatments in the sample. Models 2–3 include only observations for people who saw treatments 1–4.

	All	T1-4	PW T1-4
	(1)	(2)	(3)
T.1	0.346* (0.147)		
T.2	1.501*** (0.154)	1.154*** (0.155)	1.401*** (0.367)
T.3	1.409*** (0.136)	1.062*** (0.142)	1.099*** (0.314)
T.4	0.709*** (0.141)	0.363* (0.147)	0.678* (0.312)
Constant	– 1.199*** (0.107)	– 0.852*** (0.110)	0.511* (0.257)
Observations	2,082	1,672	571
R ²	0.103	0.070	0.045
χ ²	165.199*** (df = 4)	90.443*** (df = 3)	16.442*** (df = 3)

Note: *p < 0.05; **p < 0.01; ***p < 0.001.

²⁷ An unintended programming error in the generation of the treatments implied there were no Programmed Withdrawals decisions allocated the Control layout and therefore, if the full sample was included this would bias the treatment effects estimation. Furthermore, the comparison is done on maximal offer selection and not mean rewards because the variance in Programmed withdrawal offers is substantively lower than in the other three schemes and there are only six companies that provide the scheme, relative to an average of 12 companies for the other schemes.

substantively equivalent and is within the margins of error for Treatments 2 through 3. These findings suggest that the loss frame on its own contributes significantly to improving the choice of pension providers.

Treatment heterogeneity

One of our expectations is that the treatment effects of these design enhancements could vary in the population. In particular, we suggested that their benefits might be more pronounced for individuals with lower levels of facility with financial decision-making. We identified three variables that measure this trait in the population: levels of financial literacy, income and gender. These measures were included in the model estimating treatment effects. Of interest here is whether the average treatment effect conditional on any one of these three variables is consistent with this conjecture regarding facility with financial decision-making.

To identify heterogeneous treatment effects we adopt a machine learn approach (Green and Kern 2012; Imai and Strauss 2011; Athey and Imbens 2017; Wager and Athey 2018; Grimmer, Messing, and Westwood 2017). We estimate treatment effect heterogeneity following a Bayesian Additive Regression Trees (BART) strategy with 1100 iterations (Green and Kern 2012; Hill 2011; Duch et al. 2020). This approach to estimating conditional average treat effects (CATEs) imposes no functional form on the estimated treatment and covariate effects.

Given the earlier results and to facilitate estimation, we collapse the Control and Treatment 1 into the control condition and collapse Treatment 2 through 4 into the treatment condition. The ATE here is the difference between these two redefined treatment categories. Of interest is the ATE conditional on covariate values in the sample – the Conditional Average Treatment Effect (CATE). The number of respondents populating any one unique covariate/treatment value will be relatively small. With so few observations sharing any one of these unique covariate values, estimated differences in CATEs are likely to be driven by random variation in the small samples (Athey and Imbens 2017; Grimmer, Messing, and Westwood 2017). The challenge then is to estimate

heterogeneous effects that distinguish systematic responses from differences that are the result of chance random assignment. To this end we employ BART. Hence, we estimate CATEs for respondents sharing particular values on all combinations of relevant treatments and covariates.

Figure 10 presents the distribution of CATEs organized by magnitude along with the histogram of covariate profiles. The overall CATE (Treatments 2 through 4 versus Control and Treatment 1) is 477 and the range over all the covariate values is 333 to 559. The distribution of CATEs generated by BART suggests that all CATEs are positive, consistent with the results presented in Tables 4–5.

The lower part of Figure 10 characterizes the density of financial literacy types amongst respondents with CATEs organized along the horizontal scale shown in the upper part of the figure.²⁸ At the extreme left are respondents with the lowest CATEs, i.e. those for whom the design enhancements had the least effect on their choice of provider. Note that those with higher financial literacy (either medium or high levels) tend to be concentrated on the low end of the CATE scale. Moreover, as we move towards the high end of the CATE scale we find most of the respondents with no, or little, financial literacy.

These findings are in compliance with our expectations: the positive effects on respondents of reducing information demands (in particular eliminating the risk information from the comparisons) and adopting a loss frame were particularly evident for respondents with low levels of financial literacy. We conducted similar analyses for gender and income, but do not find any substantive or significant treatment effect heterogeneity. The results for those analyses can be found in figures A.9 and A.10.

Pension providers selected by the Chilean population

Our goal was to design a decision-making experiment that resembled the choice setting faced by a typical Chilean required to select a pension provider. We also aimed to recruit respondents who resembled the Chilean population of retirement investment decision-makers. These efforts we believe would ensure that our estimated treatment effects

²⁸Financial literacy is measured through three standard questions, two of which are on composed interest rate and the standard bat and ball question – see the survey questions in the public repository for exact wording. The financial literacy levels presented in the figure corresponds with the number of correct responses, with 0 = *NoFin.Lit* and 3 = *High.Fin.Lit*.

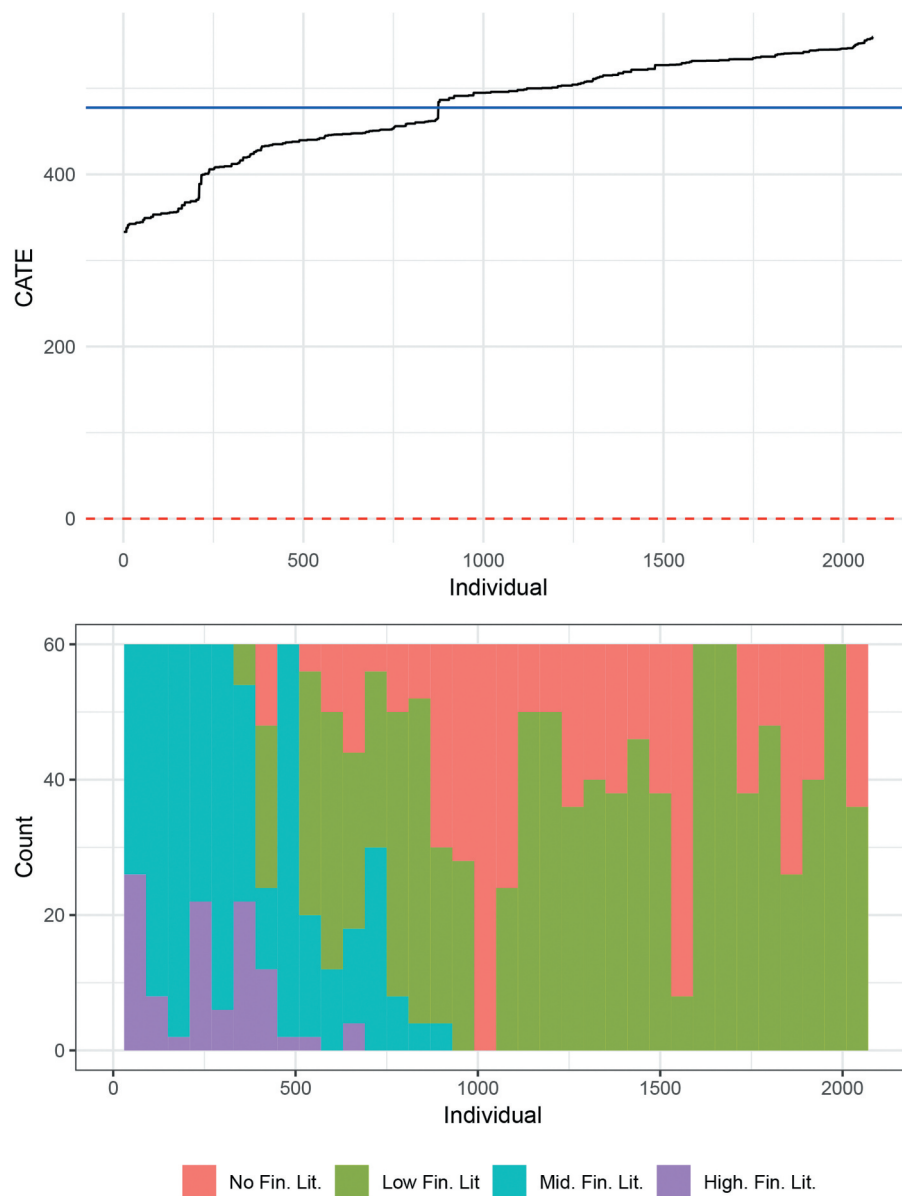


Figure 10. Financial literacy heterogeneity.

could be generalized to the relevant retirement decision-makers in the Chilean population. A comparison of the choices made by our experimental respondents with those made by real people in the Chilean population suggests the design achieved this goal. We compare the pension provider choices made in the experiment with the decisions made by the Chilean population of retirees during the time of the experiment. Figure 11 suggests that the experimental respondents made choices very similar to those made by the Chilean population. The height of the bars indicates the proportion of the top three pension provider offers selected by the decision-making groups identified on the horizontal

axis. The Control group and the Population are the most comparable here. They both made pension provider choices from identical choice sets and with the same information presentation formats (recall that the control group in the experiment received the status quo information presentation that was employed by SCOMP at the time).

In 2018–19, 47.6% of pension provider choices by Chileans were from one of the top three offers. This is very similar to the Control group in our experiment: 41% of their choices were from one of these top three offers. And as we would expect the Treatment 1 group was higher than the Population decision-makers – 53.1%. It is important to point

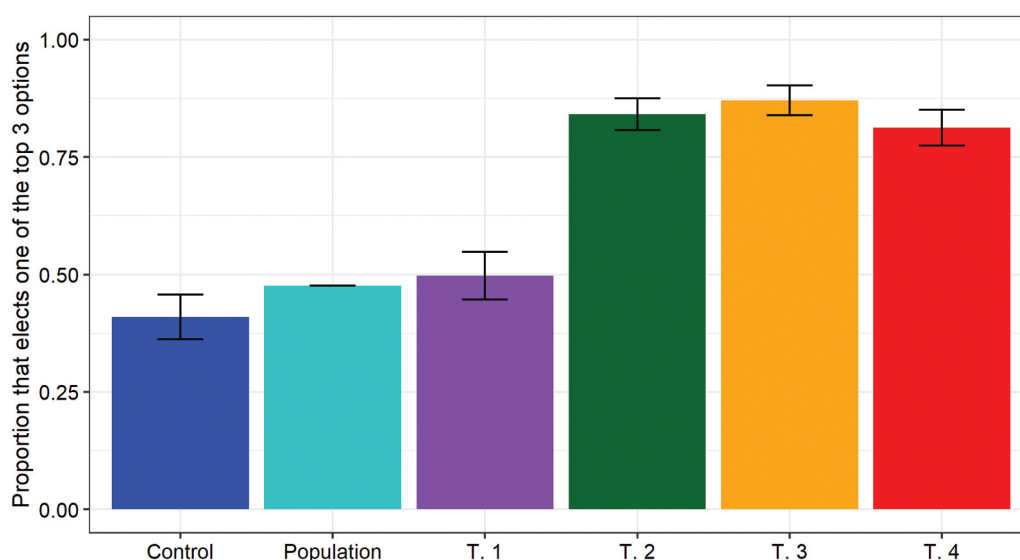


Figure 11. Proportion of top 3 offers selected by the general population between January 2018 and July 2019 in comparison to experimental outcomes. The Population bar does not have a confidence interval, as it is the effective mean of the population of interest and not a sample (source: *Superintendencia de Pensiones*).

out that we should expect the Population group to make somewhat better decisions than the Control group in the experiment because of decision-making aids. Many in the Population group received help from an independent pension advisor or from an insurance company sales agent. And, of course, those that self-selected to make unassisted decisions were not randomly assigned to this status.²⁹ The rough decision-making equivalence between our Control group and the relevant decision-makers in the Chilean Population gives us considerable confidence in our claim that improvement in the layout of the SCOMP pension provider offers would generate large welfare gains for Chileans.

IV. Discussion

The Organization of Economic Co-operation and Development (2019) reports that funded and private pensions in the world exceed \$US 40 Trillion. They also highlight the growing trend towards defined contribution pension plans; many of which require the very high stakes decision-making by consumers that we focus on in this essay. Enhancements in choice architecture that improve retirement product decisions can make significant improvements in the

returns for the average consumer and generate enormous enhancements in overall consumer welfare.

The unique design of our experiment allows us to calibrate the impact on consumer welfare of specific choice architecture enhancements in how the performance of these products is communicated. We recruited relevant segments of the Chilean population to participate in the field experiment – those yet retired 55–70 years old individuals who would most likely need to make these high stakes decisions regarding their retirement funds. By working with the Chilean Superintendencia de Pensiones, the information content of the treatments matched precisely what a typical contributor would receive if they requested performance metrics from the SCOMP certificate. The product information treatments that respondents received were customized to their gender and income demographics. And of course, we incentivized the choices they were asked to make.

We treat respondents by modifying the status quo information ‘package’ (that exactly replicates what is currently provided by the regulator). Two design choices clearly enhanced consumer welfare in this experiment: Reducing the amount of information appears to have improved choice – specifically eliminating the visibility of risk profiles of

²⁹In fact, given access to these decision-making aids, it might be surprising that less than 50% of the general population choose a top 3 offer. This occurs, however, because only 10% of those that received ‘help’ from a sales agent choose one of the top 3 offers! Source: *Superintendencia de Pensiones*.

providers leads to better decisions by our respondents. Second, adopting a loss frame also resulted in respondents selecting providers that generated higher returns. An unexpected null finding was that presenting the results in graphical format did not lead to better decision-making. Finally, as others have demonstrated, these gains to choice architecture were considerably higher for those with lower levels of financial literacy.

The results presented in Figure 9 nicely summarize the insights we provide regulators of defined contribution pension schemes. In the control condition, which was the Chilean status quo until December 2019, three-quarters of the respondents do *not* select pension offers with the highest Net Present Value – in line with the FNE (2018) report. Our results suggest that by implementing two modifications to how products are benchmarked against each other would double the number of consumers selecting the best products. Eliminating one piece of information from the comparison, in this case risk ratings, and adopting a loss frame for the performance metric resulted in half the respondents selecting the product with the highest New Present Value. Even in a relatively small country like Chile, this implies quite substantial overall monetary benefits for consumers. This is of course one online field experiment and as we have pointed out elsewhere (Duch et al. 2020) for a variety of reasons we should treat it as suggestive but hardly definitive. A more definitive conclusion regarding the impact of this choice architecture will require further research, part of which is possible due to changes in the regulation of the SCOMP certificate in January 2020 that transformed Treatment 2 into the new *status quo*.

These choice architecture results contribute to a broader public policy challenge. Our results quantify the benefits of improving choice architecture for consumers of defined contribution pension schemes. This directly concerns regulators of private and semi-private pension systems that have been adopted by a number of countries. A broader concern though is that average consumers are increasingly engaged in assessing diverse performance metrics associated with both retirement and non-retirement investment products. Alongside net present value, consumers are increasingly concerned with the extent to which

investment vehicles incorporate environmental, social and governance (ESG) criteria in their investing decisions (Friede, Busch, and Bassen 2015). As the Dutch study by Apostolakis, Kraanen, and van Dijk (2016) indicates, consumers will find it challenging to incorporate social responsible investment performance in their evaluations of retirement investment products. Choice architecture will become increasingly important in the design of information provision that helps consumers incorporate diverse criteria in their selection of appropriate retirement investment products.

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